



LAND OF THE CURIOUS



EVENT AND DATE

LECTURE 6 – CONCEPT DEVELOPMENT

CS39A0040 Product & Service Development

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AGENDA FOR LECTURE

- Lecture 6 – Concept Development
 - » Product Management Viewpoint
 - » Product Development Team Viewpoint

”

” The difficulty lies not so much in the in
developing new ideas as in escaping
from old ones”

John Maynard Keynes

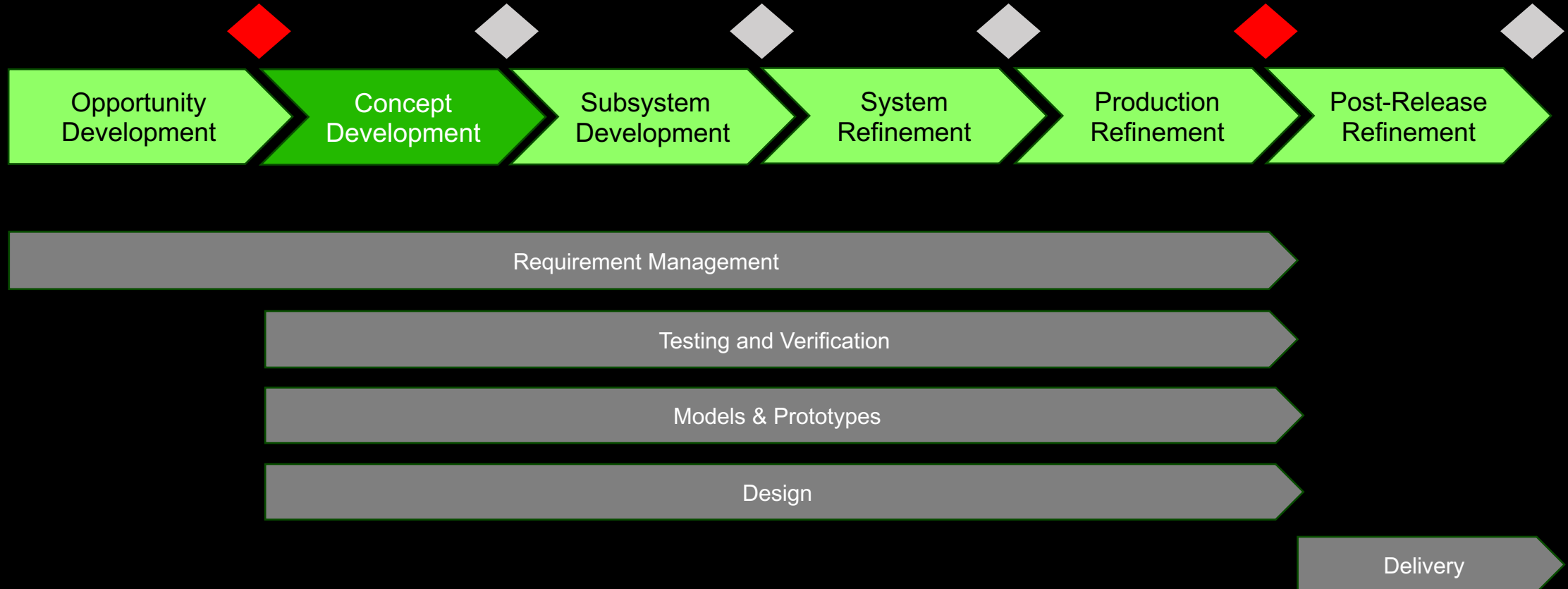




CONCEPT DEVELOPMENT - THE BIG PICTURE

Different views of concept development

PRODUCT DEVELOPMENT



PRODUCT CONCEPT EXAMPLES



*USAF NGAD 6
Generation Concept*



*Porsche 919 Street
Concept*



*Mayer Cruise Ship
Concept*

PRODUCT CONCEPT: INNOVATIONS FAILURES



*DeLorean DMC-12
(1981 – 1983)*

*Performance & safety
issues*

9000+ built



*Apple Pippin
(1995)*

*Too many
requirements*

c. 100,000 sold



*Google Glasses
(2013 - 2015)*

*Privacy concern,
bugs, battery life*

Price

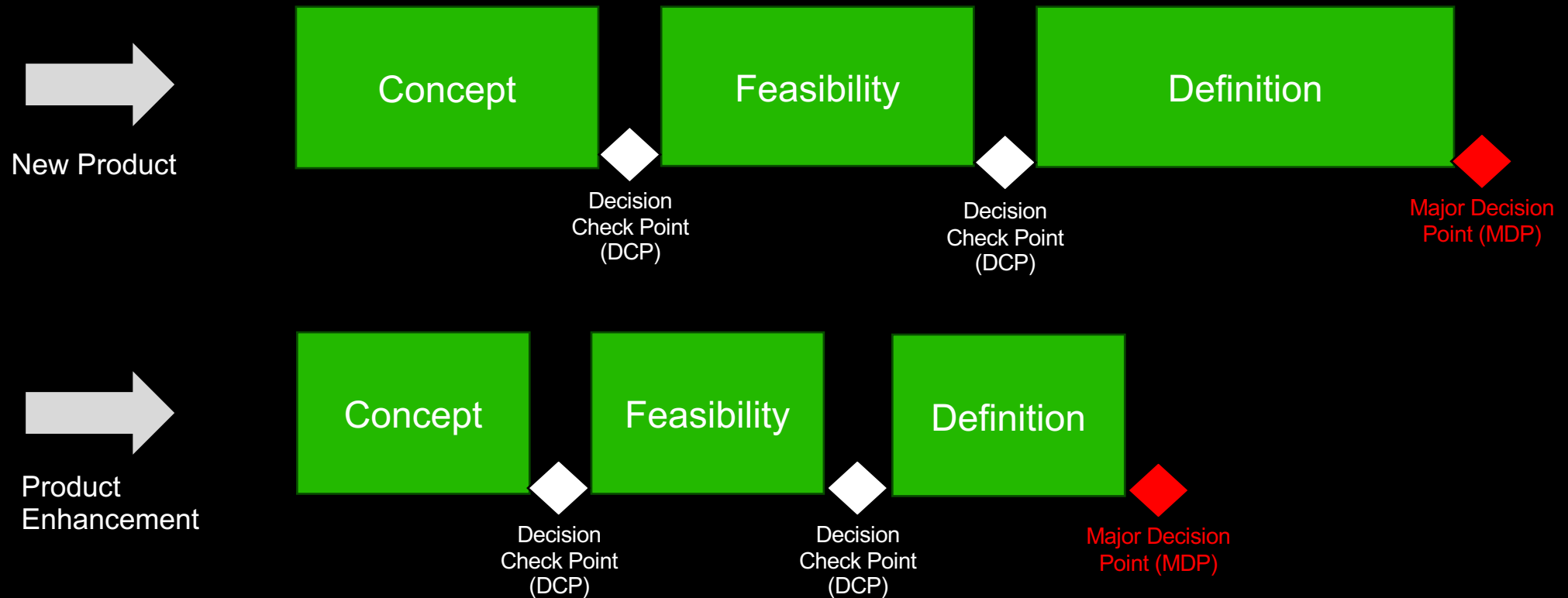


CONCEPT, FEASIBILITY & DEFINITION PHASES

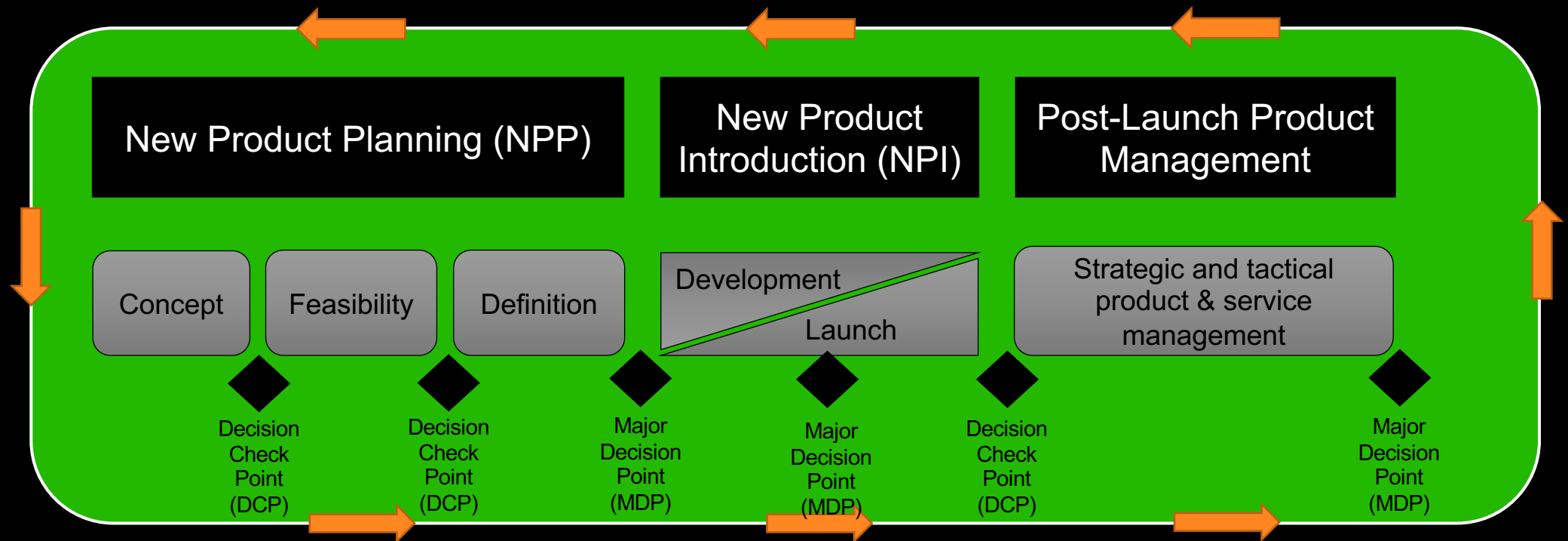
Methods to define the product & service concept & feasibility for Product Development & Product Mangement

Product Management View on Phases

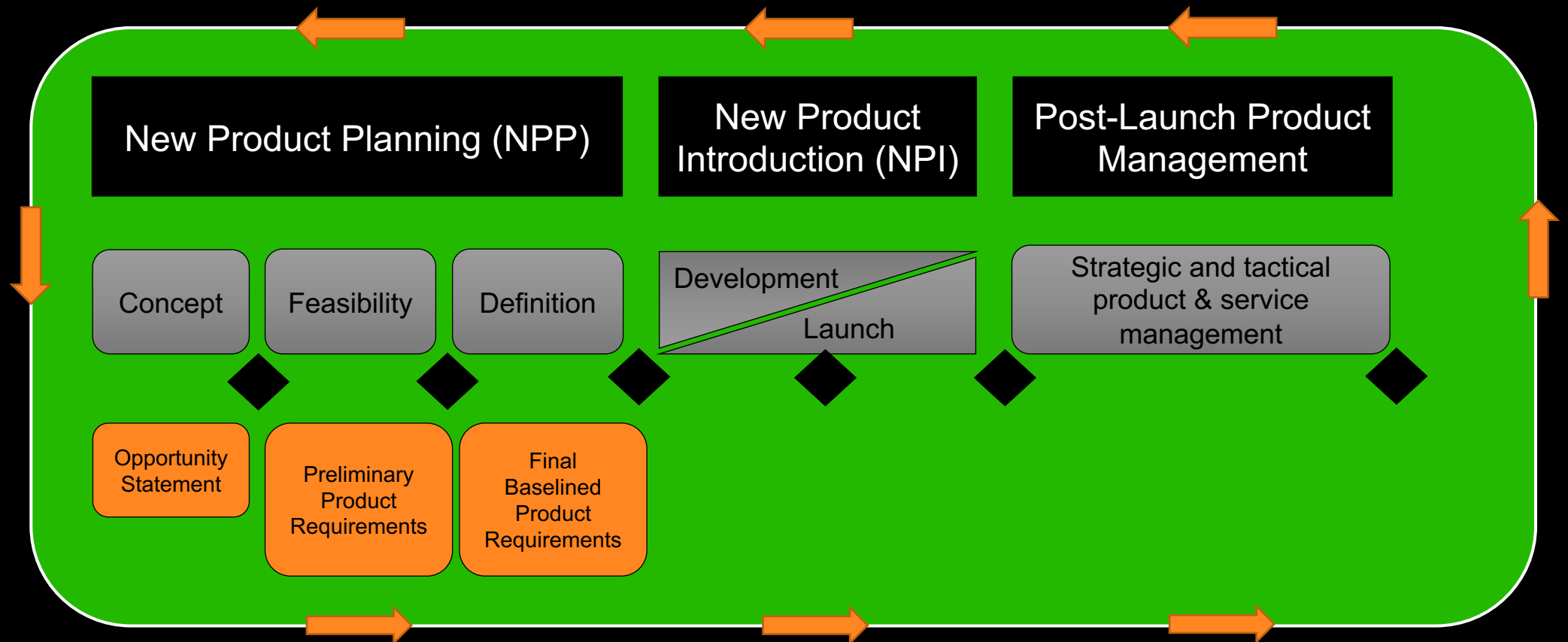
EXAMPLES OF THE PLANNING PHASE



GENERIC PLANNING & DEVELOPMENT PROCESS



REQUIREMENT MANAGEMENT



■ Making a molehill out of a mountain

PRODUCT MANAGEMENT : CONCEPT PHASE

- » The concept phase is a fast-paced business decision making process
- » New product concepts come from many sources
 - Market Analysis
 - Observation
 - Structured ideation activities
- There are more ideas than there are resources to commercialise them into a product or service
 - Not all ideas make good business sense
 - Structure process helps to narrow the field quickly and efficiently
- Product concepts must be in perspective and need a solid strategic plan

Is there really a business case?

PRODUCT MANAGEMENT : FEASIBILITY PHASE

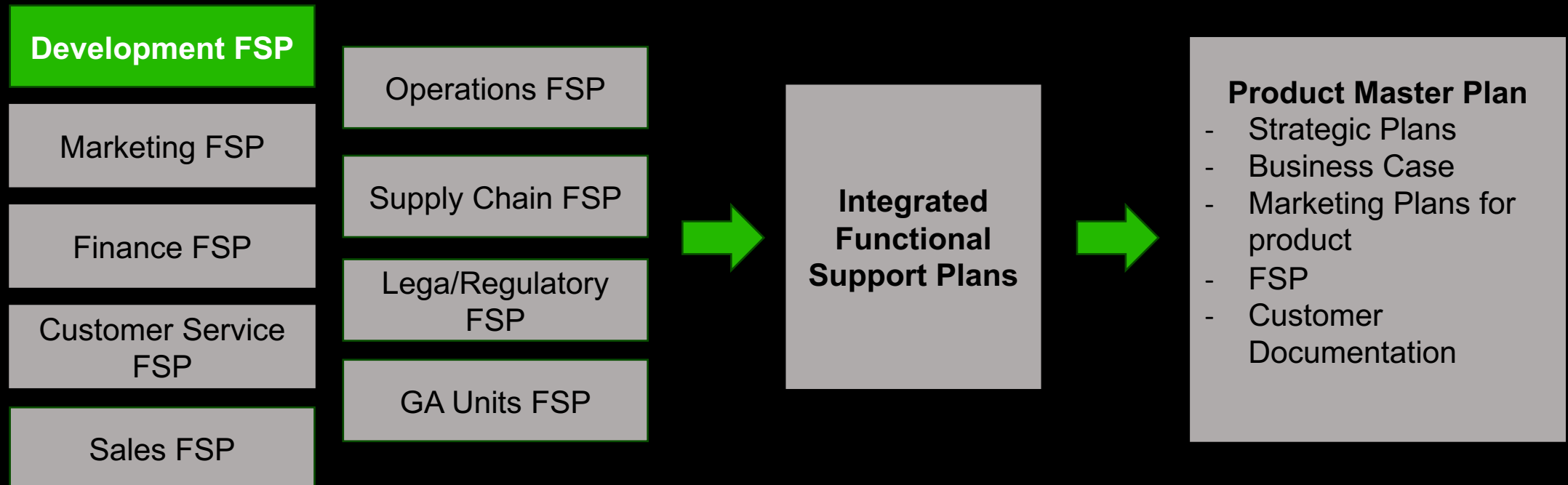
- » The cross-functional product team decide if there is realistic busienss to justify contuniuing with development project
- » Product Manager align the cross-functional team to ensure the product can be developed, launched and sustained on the market
- » Key information (documents)
 - Business Case
 - Product Marketing Plan
 - Product Requirements
 - Product Launch Plan
- In the feasibility review, key business questions about the product must be understood

■ Appearance are everything

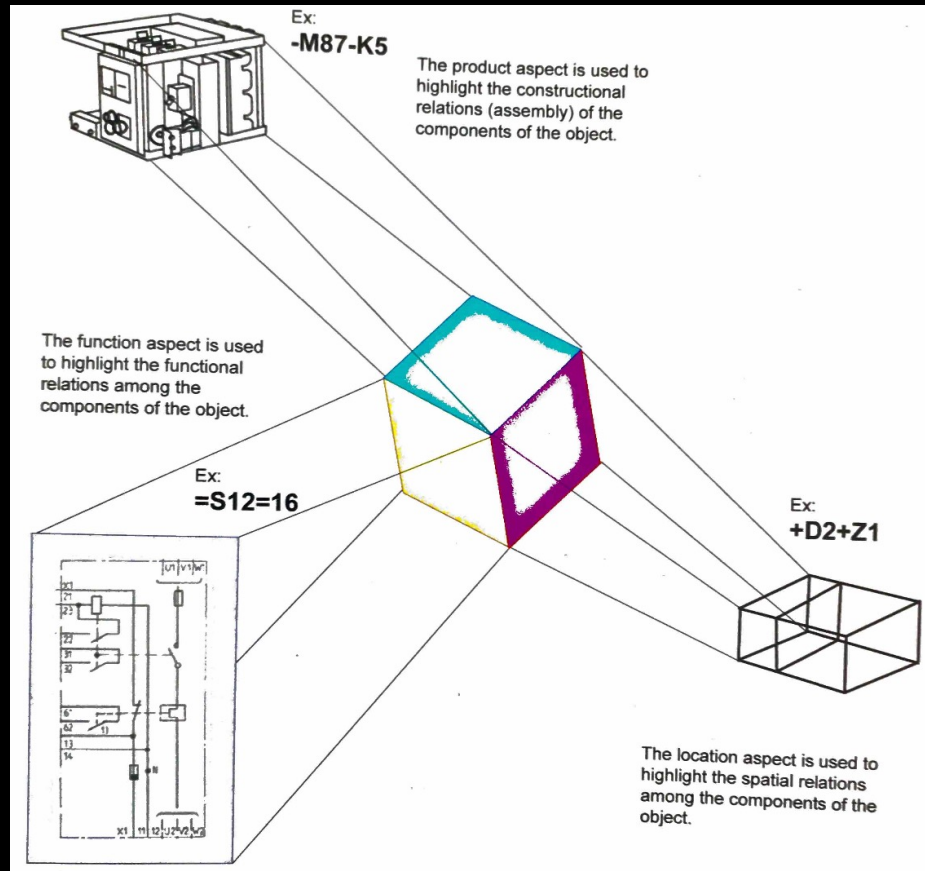
PRODUCT CONCEPT : DEFINING THE PRODUCT CONCEPT

- » Solid definition of the product from the customer's perspective improves the product's chances of achieving the market goals
- » The Definition Phase:
 - The MOST important linkage between the needs of the customers with the design & capabilities of the product.
 - Clarifies the product's content
 - Completion of all planning information (documentation) including Business Case & Product Launch Plan

PRODUCT MASTER PLAN

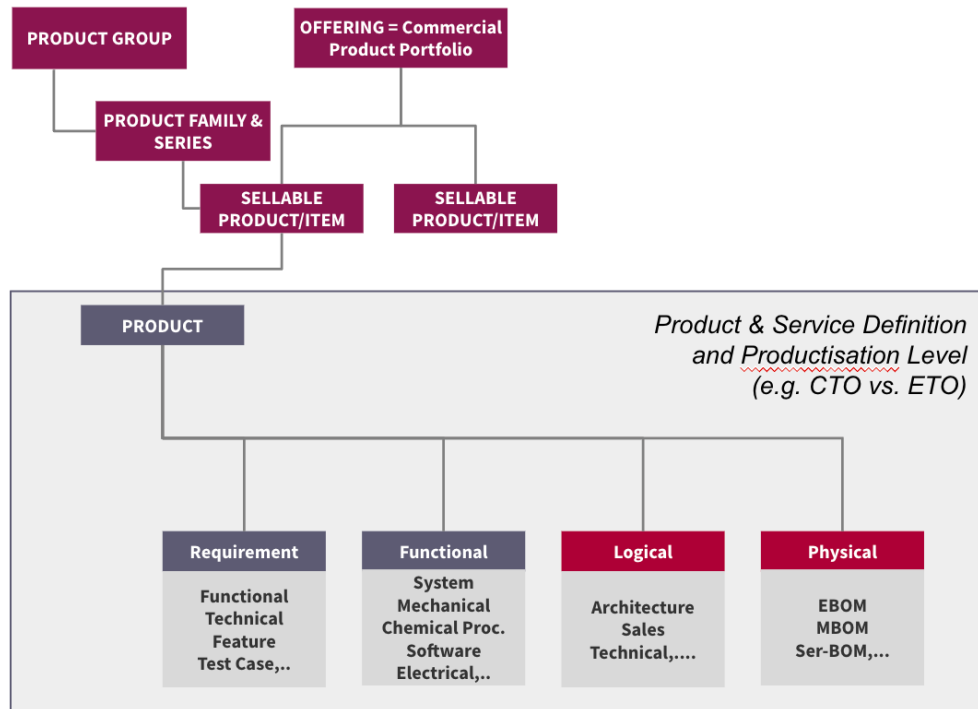


TYPICAL PRODUCT DEFINITION VIEWS



- The physical aspect is used to highlight the constructional relations of the component of the object.
- The functional aspect is used to highlight the functional relations among the component of the object.
- The location aspect is used to highlight the spatial relations among the components of the object.
- There can be more product & service viewpoints.
- Avoid object overloading.

PRODUCT DEFINITION VIEWS



- Products and service can have multiple types of definition views (RFLP).
- Sellable product definition is granular compared to technical and functional definitions
- Physical item and structures are not optimal when defining complex sales, technical and functional instances.
- Product and service definition is the detail level that the product is defined in the product-service portfolio.
- Product, structures and its components can have location tag associated with e.g. PCB reference designator or pump-line tag.

EXAMPLE : FUNCTIONAL MODEL & FEATURES

B. Nepal et al. / Int. J. Production Economics 96 (2005) 157–174

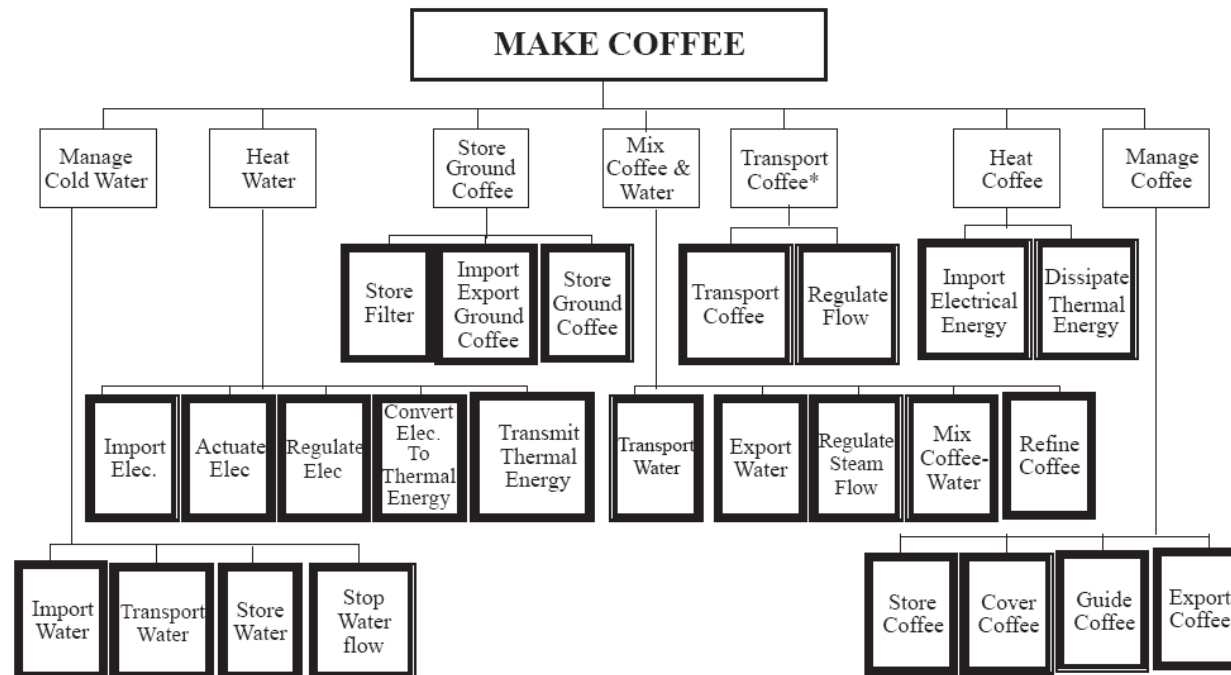


Fig. 8. Functional decomposition of a typical coffeemaker.

- Defines how the product functions
- Defines the products features
- Functional structure is key for a real-time simulation based Digital Twin

EXAMPLE : PHYSICAL PRODUCT MODEL & TECHNICAL FEATURES

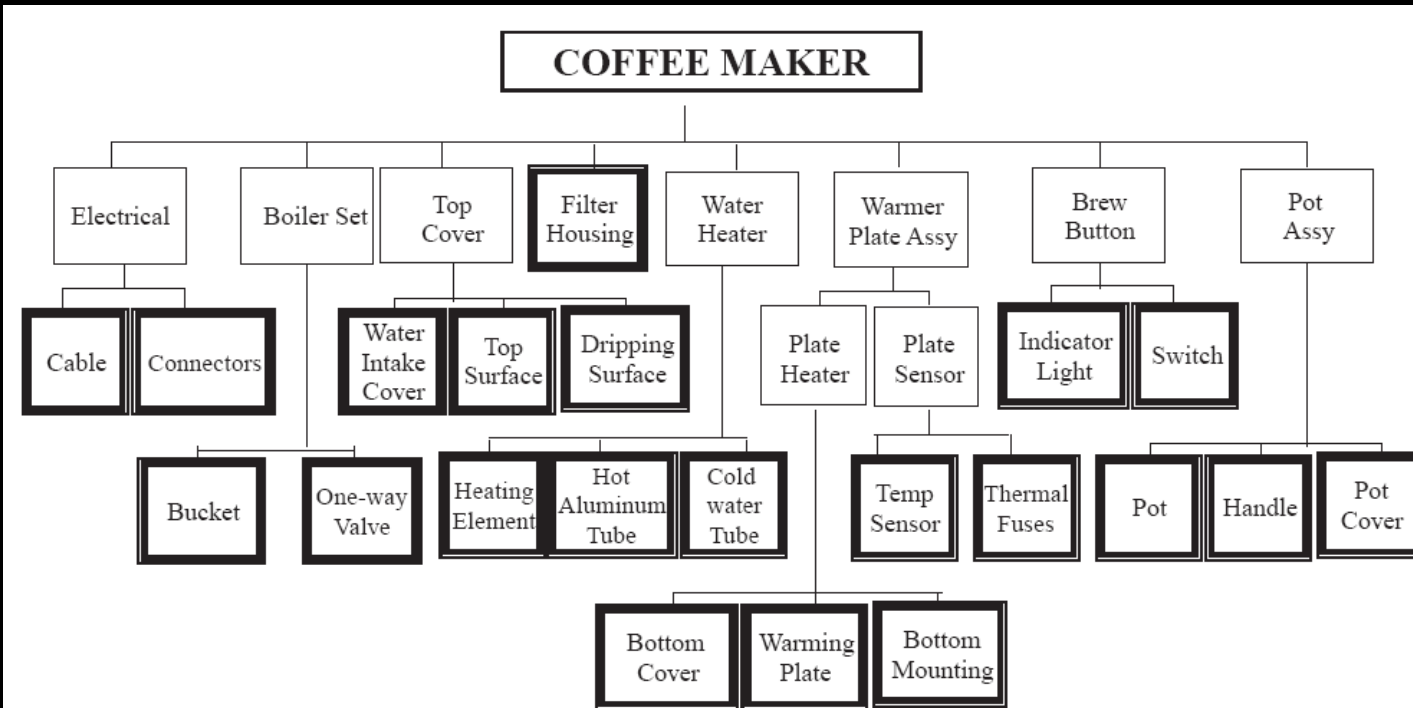


Fig. 9. Physical decomposition of a generic coffeemaker.

- Defines how the product looks
- Defines the products technical features
- Defines the manufacturing breakdown
- Physical structure is key for a real-time simulation based Digital Twin (it is what we drive in virtual world)



PRODUCT CONCEPT DEVELOPMENT PHASE

Methods to define the product & service concept



PRODUCT CONCEPT : OBJECTIVE PRODUCT CONCEPT PHASE

KEY CONCEPT DEFINITIONS

Concept is an idea solving a design need including:

- a description of the main operating principles,
- major components, and structural, logical, and operating interfaces between the components

Strong Concept is a concept that has been rated as superior to any other known concept, with specific advantages and few or no disadvantages

OBJECTIVE OF CONCEPT DEVELOPMENT

The develop team responsibility is to:

- Define & select a strong concept of from all proposed concepts
- Rough understanding of the major components or subsystems
- Define System performance
- Estimate subsystem performance

■ Evolve the information that characterizes the *solution*

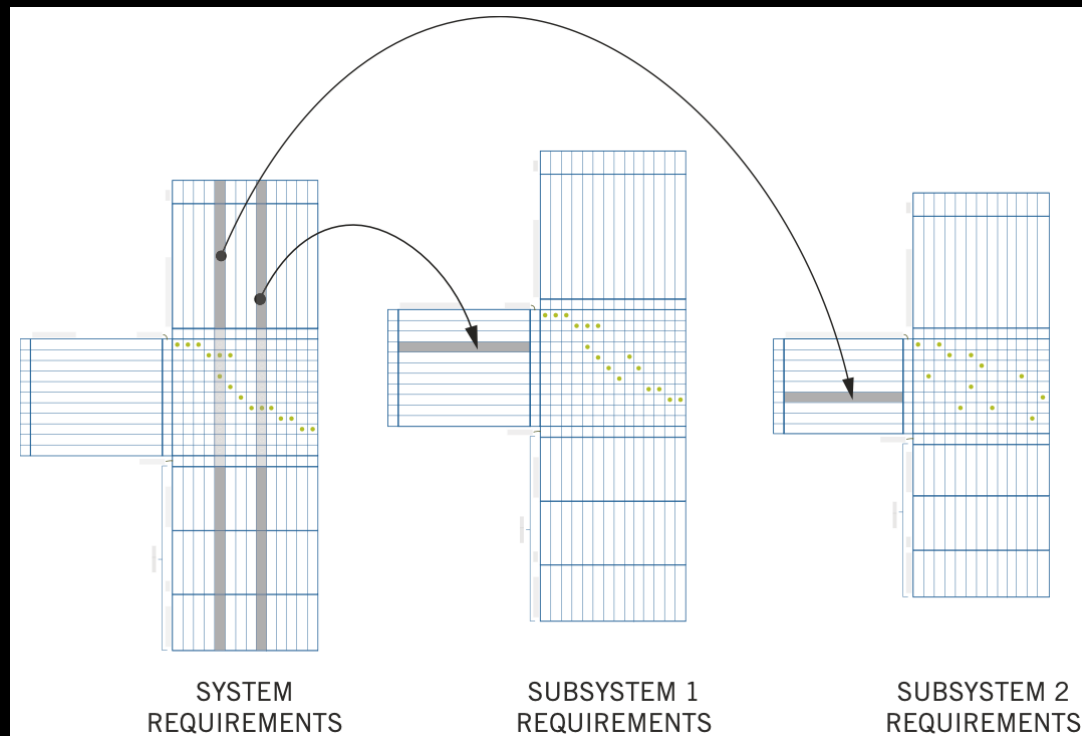
PRODUCT CONCEPT DEVELOPMENT

» The design opportunity evolves from a very general idea to a high-quality system architecture, which has the following items:

- An accurate, unambiguous, transferable definition of the selected product concept, where enough detail is provided about
 - Spatial (space allocations) & structural relationships of the principal subsystems
 - Basic cost, size, weight,
 - Feasibility estimates of the design.
 - Demonstration how the concept & its parts meet the opportunity.
- The decomposition of the system into major components or subsystems
- The definition of the interfaces between the subsystems.
- Opportunity definitions for each of the subsystems (opportunity matrix sections A, B, C, D).
- Target values for the system and subsystems

What are trying to do?

SYSTEM & SUBSYSTEM REQUIREMENTS



- The Requirement Matrix aligns the system & subsystems
- Target values of system performance measures.
- Not all system performance measures apply to every subsystem.

PRODUCT CONCEPT DEVELOPMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
Requirements	Subsystem Requirements	Section A-D of the requirements matrix for each subsystem	Opportunity Development summary	Complete and appropriate.
	Target values for the system & subsystems.	Section E of the requirements matrix for the system and all subsystems	Consistent with ideal values? Achievable with the selected product concept? Target values for all performance measures? Distinction between constraints, success measures, stretch goals? Trade-offs to less-than-ideal performance justified?	Consistent with expectations of market
Tests	Justification for concept selection (model analysis and/or prototype test data) demonstrating concept feasibility	Summary & evaluation of considered concepts demonstrating feasibility of selected concept. Model and/or prototype test reports demonstrating the validity of the selected concept.	Are the test methods complete and correct? Is there enough detail for a third party to repeat the tests?	The desirability of the concept is demonstrated.
Models	Rough technical models of the product concept.	Low-fidelity fundamental models of the concept's operating principles. Statistical models describing the performance of experimental prototypes or related existing products.	Are the models reported in enough detail to allow a third party to use them?	Not directly approved; used with tests.
Prototype	Simple prototypes of the product concept.	Rough prototypes (foamboard, paper, foam, clay, cardboard, plywood, etc.) used to show how the concept functions.	Are the prototypes appropriate for the intended use?	Not directly approved; used with tests.

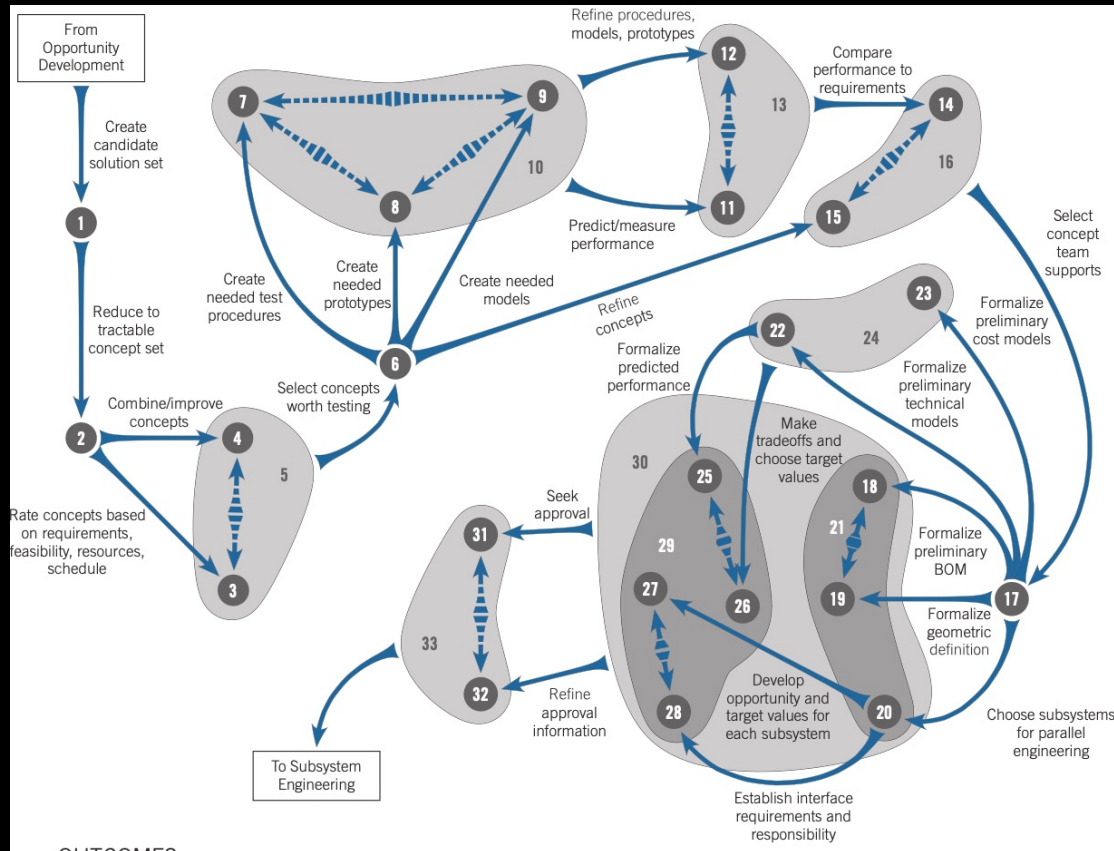
PRODUCT CONCEPT DEVELOPMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
Design	Geometric & other appropriate definition of the concept.	For example: - Annotated hand sketches of concept. - Overall system CAD model. - Layout drawing & skeleton drawing. - Notes explaining concept. - Block diagrams of electrical or fluid systems.	Is the design clearly communicated? Can a third party understand the intended design?	The design is transferable to support estimates on: - cost, size, weight - feasibility estimates.
	Decomposition of product concept into subsystems.	- Tree or other relationship diagram that shows structure of decomposition. - List of subsystems.	Are the subsystems and their relationships clearly shown?	Decomposition is appropriate for the selected concept.
	Subsystem interface definitions.	- Interface matrix showing where subsystems interact. - Product-focused requirement statements for each of the interfaces. - Performance measures for each of the interfaces. - Sub-team responsibility assignments for each interface.	Is the interface matrix complete? Are the interface definitions complete? Are the interface definitions appropriate to achieve the desired performance?	The interfaces are fully defined and appropriate for the concept.
	Preliminary Bill of Materials.	PDM / PLM lists all known components, even if they have only a part name at this time.	Is the Requirement, Functional, Logical & Physical (RFLP) structure complete at the level of known detail? BOM is engineering viewpoint	The RFLP structure & BOM are appropriate.

PRODUCT CONCEPT DEVELOPMENT PHASE

	Requirement Information	Artifacts	Checking Criteria	Approval Criteria
	Tools:	Definition of product: - RFLP structures & Bill of Materials (e.g. Engineering BOM) - Requirements matrix, scoring matrix, screening matrix, - Sketching, 3D models & drawings Tools to define & validate the product: - Bio-inspired design, brainstorming, competitive benchmarking, controlled convergence, decomposition, - Internet research, interviews, literature review, method 635, mind maps - Prototyping, recombination table - Theory of inventive problem solving (TRIZ).		
	Common Pitfalls:	- Concept fixation; - premature concept critique; - reinventing the wheel; - vague interfaces; - decision delay.		

TOP LEVEL ACTIVITY MAP



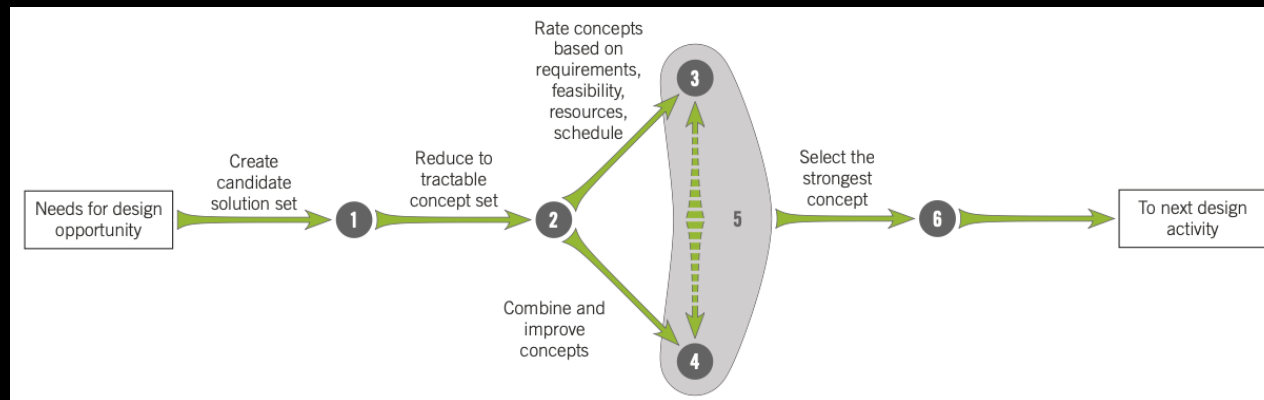
BASED ON OPPORTUNITY DEVELOPMENT PHASE

OUTCOMES FROM CONCEPT PHASE (highlights):

- Candidate Solution Set
- Concept Rating
- Improved Concept Set
- Tests and prototypes to evaluate concept
- Predicted / measured performance
- Final Concept
- Preliminary Bill Of Material
- Geometry definition
- Preliminary technical & cost models
- Subsystem requirements
- Interface definitions
- STAGE APPROVAL

MOVE TO SUB-SYSTEM ENGINEERING

DEVELOPING A STRONG CONCEPT



BASED ON OPPORTUNITY DEVELOPMENT PHASE

1. Candidate solution set
2. Reduced candidate solution set
3. Ratings for each candidate
4. Improved concept set
5. Set of rated concepts
6. Selected concept

Move to next design activity



CREATING CANDIDATE SOLUTIONS

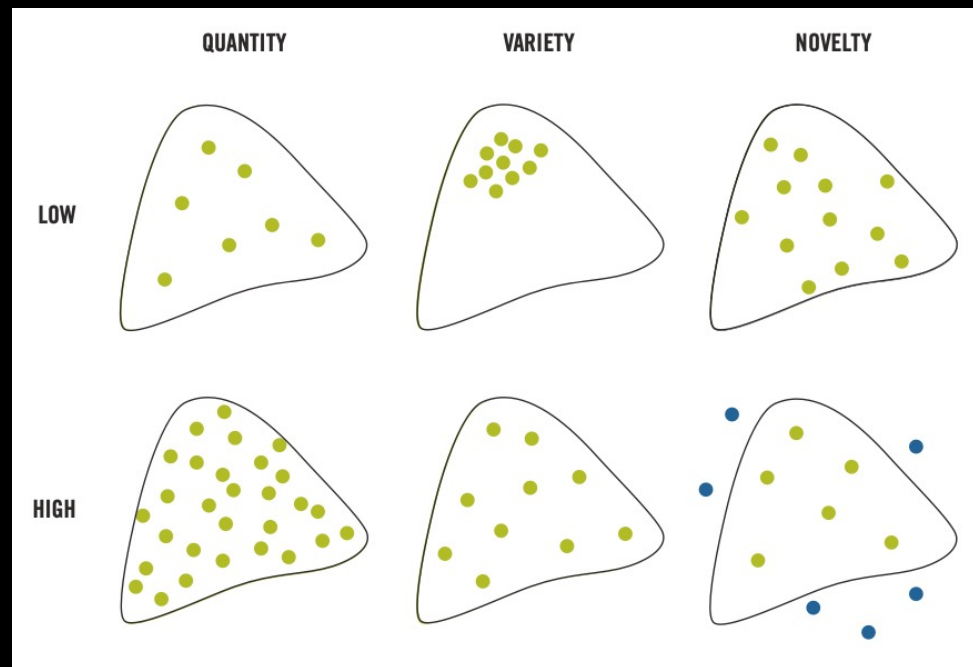
The four attributes of a concept set evaluated to assess its appropriateness (Shah et al) for critical and mundane parts:

1. Concept quantity
2. Concept variety
3. Concept novelty
4. Concept quality

Challenge of the Product Concept Development Phase are:

1. Determining the best possible design is
2. Recognize that the *best* design can only be identified in the context of *multiple* designs.
3. Essential to create a set of potential solutions, rather than just a single solution

EXAMPLE: CREATING CANDIDATE SOLUTIONS



Variable impacting solution set adequacy.

Black lines are borders of the known design space
Dots are candidate solutions.

1. Candidate set quantity is the number of distinct candidate solutions.
2. Variety is an indication of how broadly the design space has been explored.
3. Novelty is a measure of how the design space has been expanded.
4. Quality evaluates how well candidate solutions meet the design needs

Refer to Chapter 11.6

CONCEPT SCREENING MATRIX

	Concept 1	Concept 2	Reference	...	Concept M
Requirement 1	+	+	=		+
Requirement 2	+	-	=		+
Requirement 3	=	-	=		=
...					
Requirement N	-	-	=		=
Number of +'s	2	1	0		2
Number of ='s	1	0	4		2
Number of -'s	1	3	0		0
Net Score	1	-2	0		2
Improve/Combine?	Improve	Combine	Combine		None

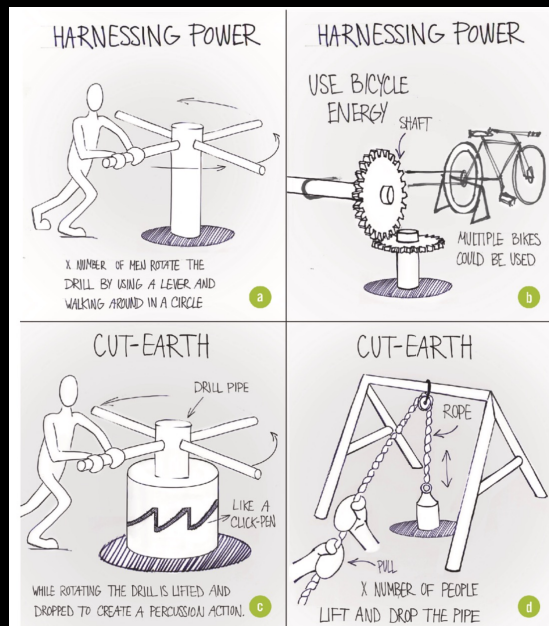
Concept Screening Matrix.

The Concept Screening Matrix is:

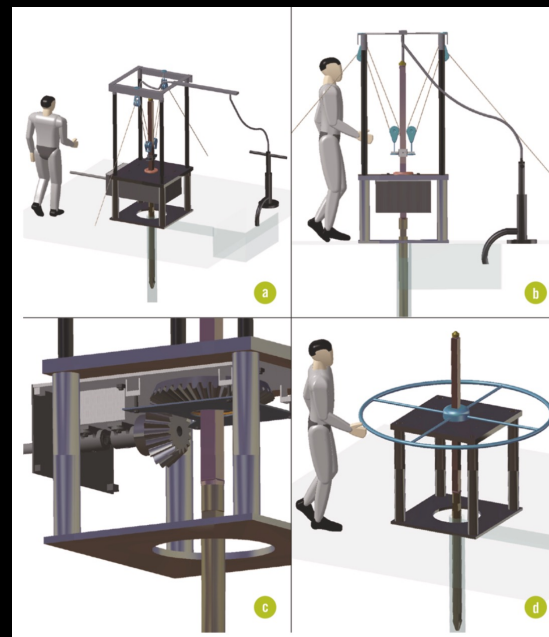
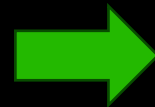
- Rough review at various candidate concepts.
- Rates proposed concepts.
- Compare to a **reference concept** (benchmark design or concept).
- Quick and effective way to reduce the set of concepts

Refer to the Book Chapter 5.2

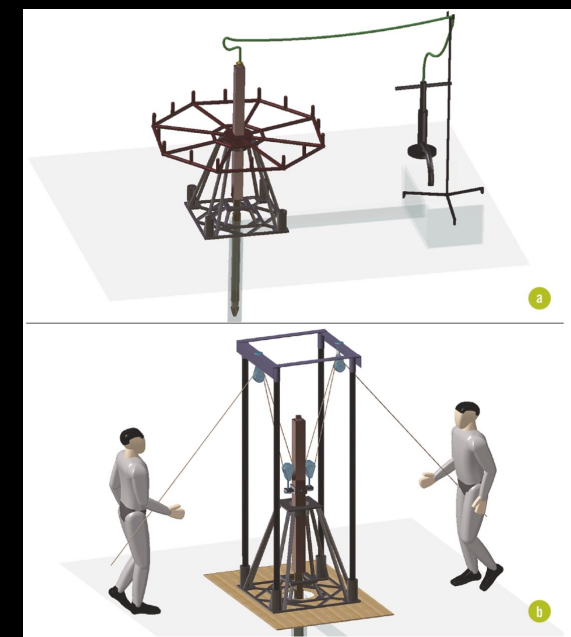
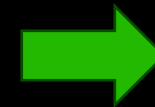
EXAMPLE : HUMAN-POWERED WATER WELL DRILL



Early Stage Concept

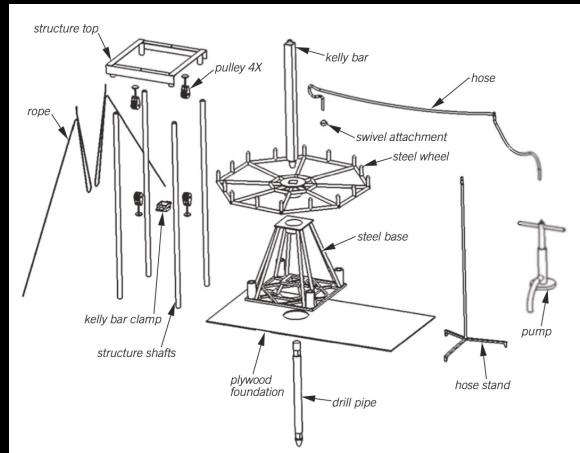


Mid Stage Concept

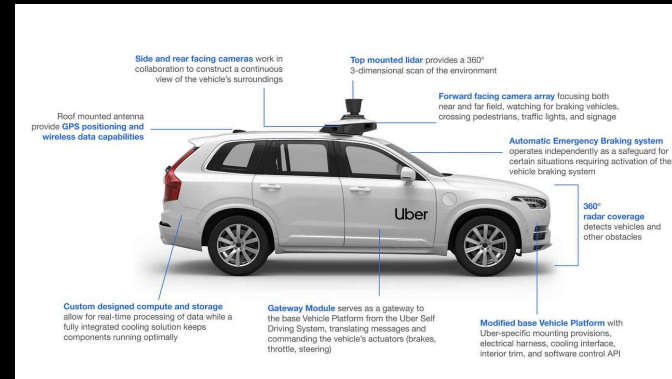


Late Stage Concept

SELECTED CONCEPT : SUBSYSTEMS & INTERFACES

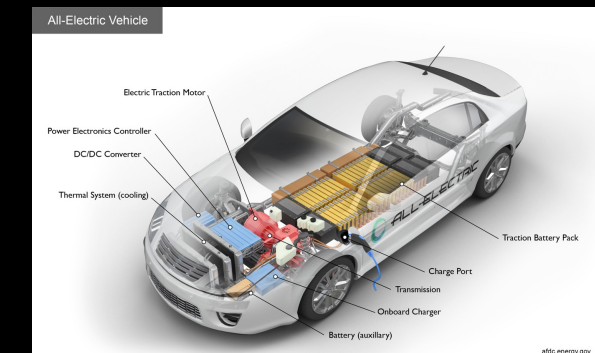
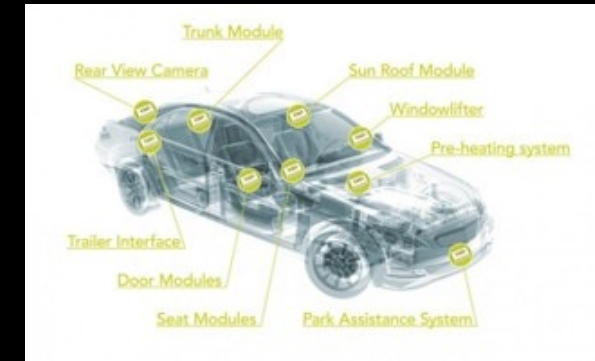


Cocnpet: System & Subsystems



System Interfaces

Subsystem Modules



PRODUCT MODULARITY

Variations

Customized products

- Designed and manufactured on order

Modularized products

- Mass customization
- Ready-made designs & manufactured on order

Standardized products

- Mass production
- Designed and manufactured to stock

- Manufacturing industry balances between customisation & modularisation
- Standardisation can be done on e.g. component and service level

Production



ANY
QUESTIONS
?

